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Business Unit: South African Environmental Observation Network (SAEON)

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Tool: Air Quality PM2.5 Dashboard

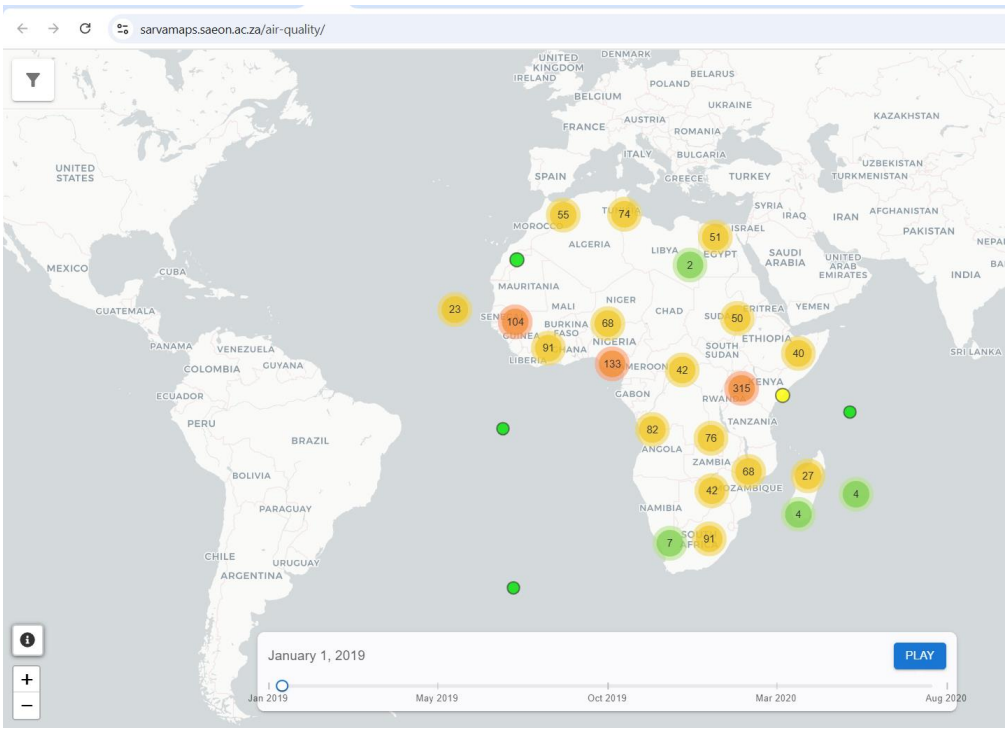
Public Link: https://sarvamaps.saeon.ac.za/air-quality/

Private Link: http://192.168.118.18/air-quality/

GitHub: https://github.com/nrf-samkelo/sarva-air-quality-web

1. Overview

The air quality dashboard provides historical time-series PM2.5 data across African countries. It replaces its former ArcGIS hosted air-quality dashboard which was fully migrated on 08 May 2025, using open source tools, improving accessibility and costs saving for SAEON. The dataset used contains daily estimates of PM2.5 particulate matter in 1466 cities across Africa as well as an additional 496 smaller population clusters in South Africa.



2. Technical Architecture

- **Frontend:** React + Leaflet.js
- **Backend:** Express.js API
- **Database:** PostgreSQL + PostGIS
- **Deployment:** NGINX on SAEON VPN
- **Data Source**: https://catalogue.saeon.ac.za/records/10.15493/SARVA.301020-2
- **Security:** SSL via NGINX (all incoming traffic logged)
    - Certificate: /etc/letsencrypt/live/sarvamaps.saeon.ac.za/###.pem
    - Key: /etc/letsencrypt/live/sarvamaps.saeon.ac.za/####.pem

3. Features

- Interactive map of PM2.5 sensors
- Time-series drill-down for PM2.5 values
- Map Layers: Population and PM2.5
- Map Filters: City, PM2.5 level, population density

4. Operational Value

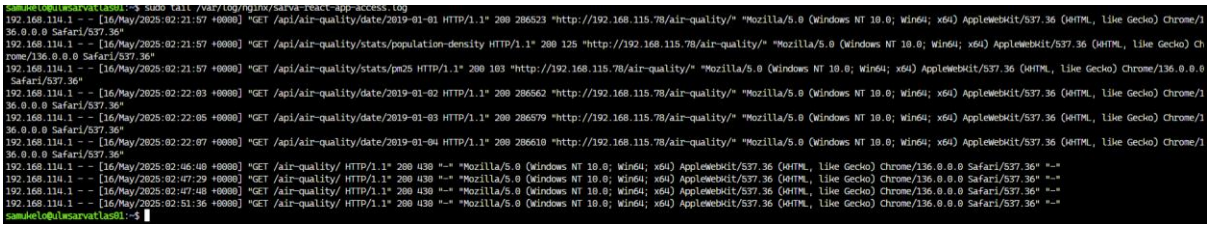
- Accessible from SARVA Atlases site: https://sarva.saeon.ac.za/atlas/#
- Audience**:** Academics, public users, environmental analysts
- Supports climate risk profiling, policy planning, pollution monitoring for major African cities

5. Future Improvements

- Enable download of timeseries data
- Add layer for NO2 and SO2

6. Feedback and Performance

- Usage stats:



- Known issues: None
- Planned stakeholder feedback: Q3 2025